Common-Source NMOS Amplifier

Portfolio Project – Analog Electronics & Circuit Simulation

Designed, simulated, biased, debugged, and documented using KiCad and ngspice.

Executive Summary

This project demonstrates the complete design workflow of a Common-Source NMOS amplifier. The objective was to design a transistor amplifier, establish a stable DC operating point, simulate circuit behavior using ngspice, and verify node voltages through transient and operating-point analysis. The project required circuit debugging, bias-network design, waveform interpretation, and technical documentation.

Technical Skills Demonstrated

- Analog circuit design
- MOSFET biasing and operating-point analysis
- SPICE simulation (ngspice)
- KiCad schematic capture
- Signal conditioning concepts
- Engineering troubleshooting and validation
- Technical reporting and documentation

Final Circuit Implementation

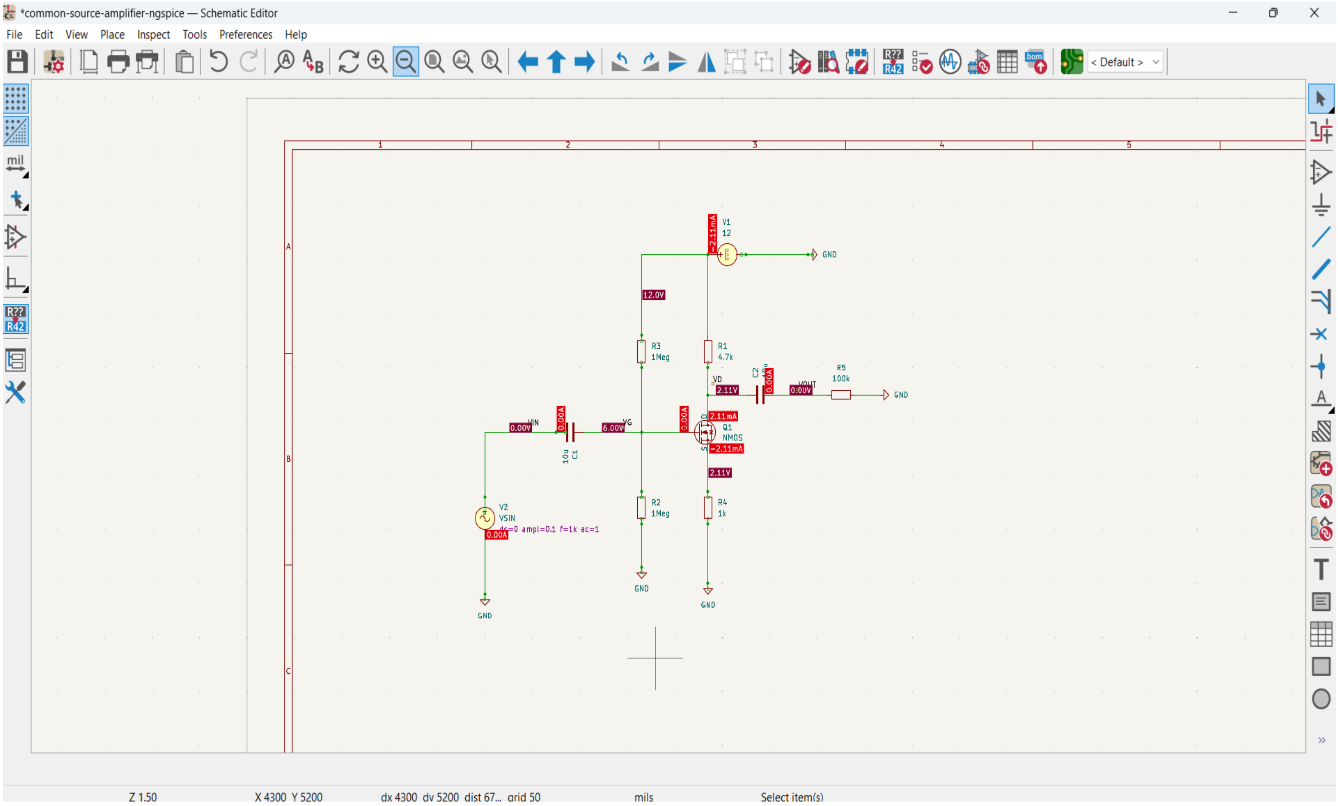


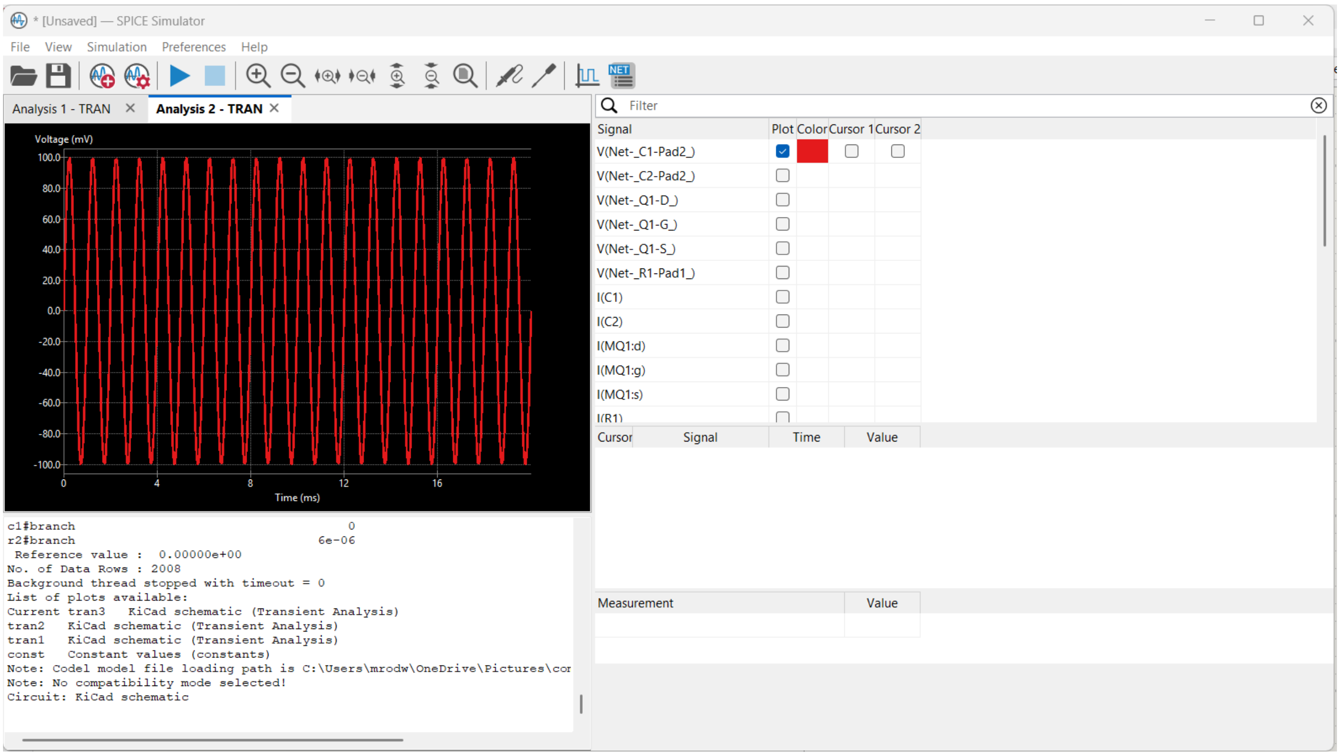
Figure 1. Final NMOS common-source amplifier. The design uses: a voltage-divider bias network (R2/R3), source degeneration resistor (R4), drain load resistor (R1), and AC coupling capacitors (C1 and C2).

Design Calculations

Supply Voltage: 12 V
Drain Resistor: 4.7 kΩ
Source Resistor: 1 kΩ
Gate Divider: 1 MΩ / 1 MΩ
Input Signal: 100 mV amplitude, 1 kHz sinusoid

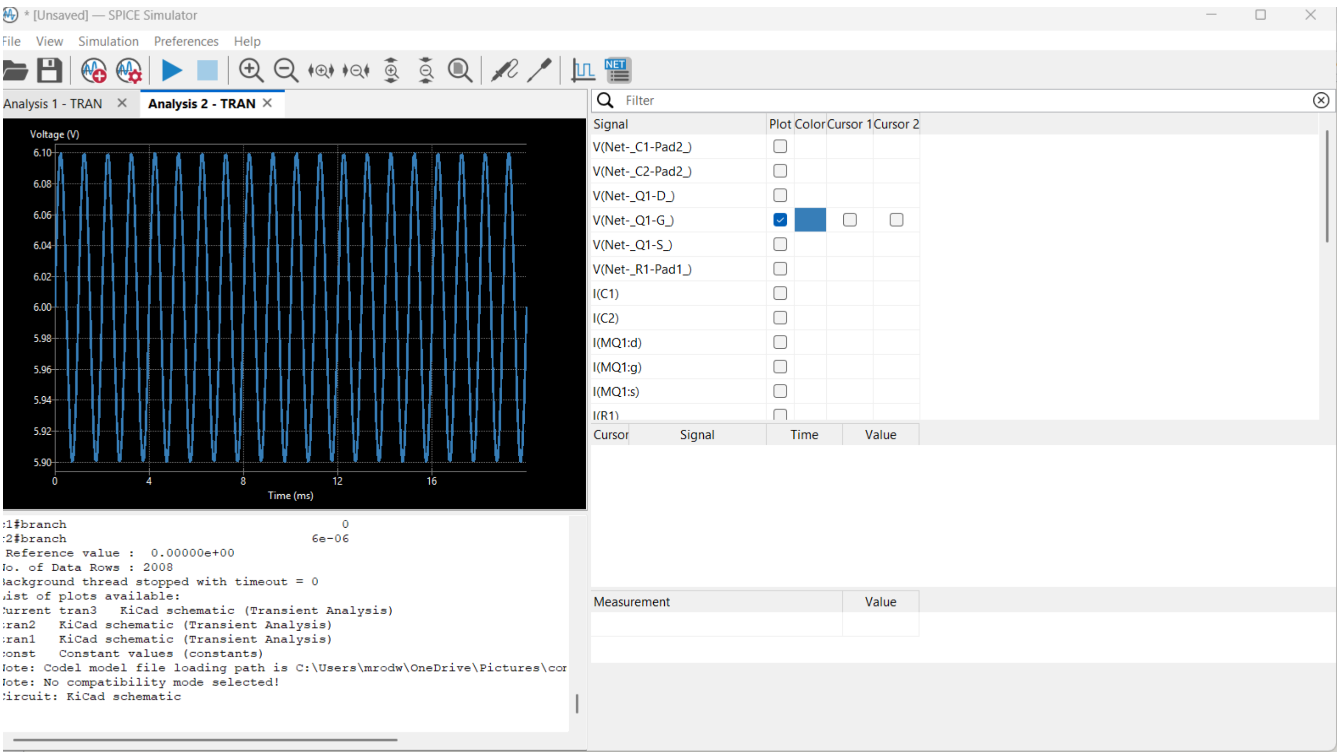
Gate bias voltage: VG = 12 × (1MΩ / (1MΩ + 1MΩ)) ≈ 6.0 V

Input Waveform (VIN)



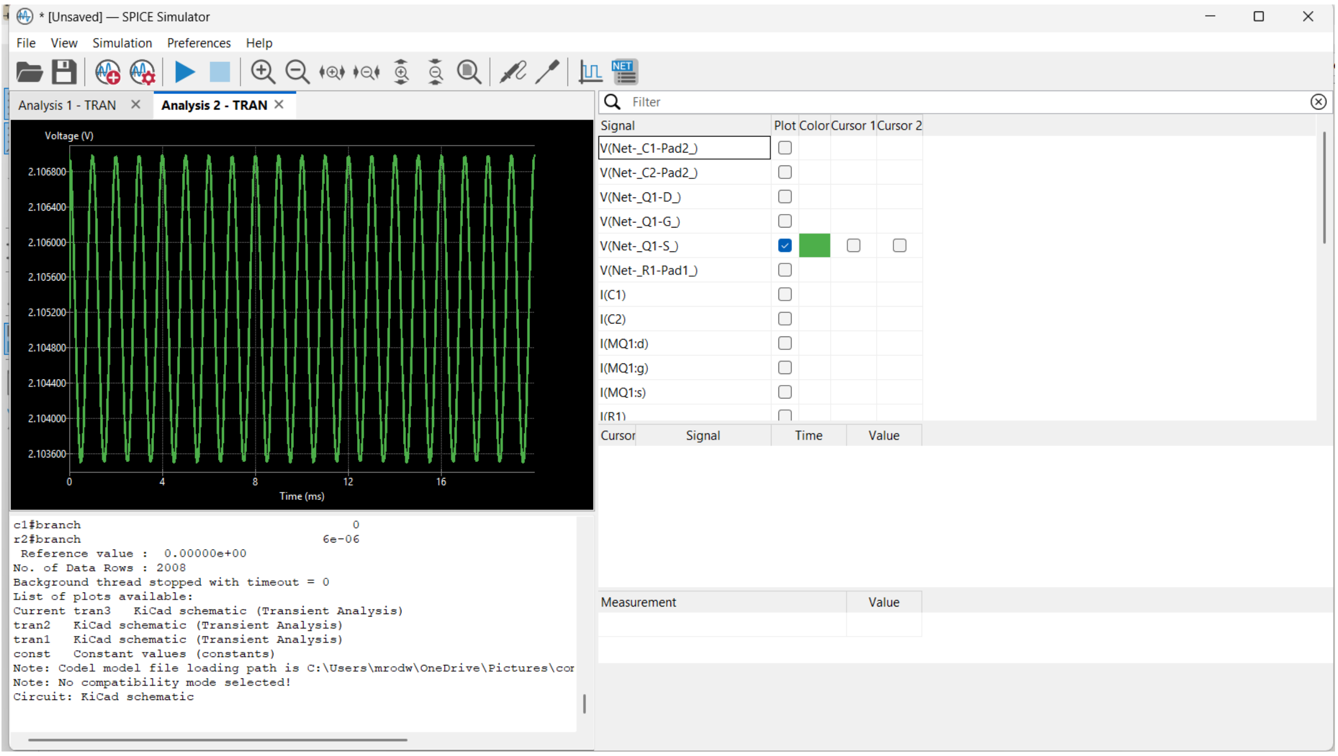
Simulation result showing input waveform (vin) during transient analysis.

Gate Voltage Response



Simulation result showing gate voltage response during transient analysis.

Source Voltage Response



Simulation result showing source voltage response during transient analysis.

Engineering Analysis

The gate voltage is established by a high-impedance voltage-divider network. This creates a predictable DC bias point while allowing the AC input signal to be coupled through capacitor C1 without disturbing the operating point. The source resistor introduces negative feedback, improving bias stability. Transient simulations confirmed that the amplifier responds to the input excitation and maintains stable operation throughout the simulation interval. Operating-point analysis showed approximately: $V_G \approx 6.0$ V, $V_S \approx 2.11$ V, $V_D \approx 2.11$ V, $I_D \approx 2.11$ mA.

Debugging & Validation

A significant portion of the project involved troubleshooting simulation configuration, signal probing, operating-point setup, waveform visualization, and verification of node behavior. The final design successfully produced repeatable simulation results and documented waveforms.

Recruiter Summary

Designed and simulated a Common-Source NMOS amplifier using KiCad/ngspice. Implemented transistor biasing networks, performed operating-point analysis, captured transient waveforms, verified circuit functionality, and documented engineering results in a professional report format.